

Paper 16.50 - Continuum Edge Residual Claim Contract

CQE-paper-16.50

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-16.50.md

Paper 16.50 - Continuum Edge Residual Claim Contract

Purpose

This contract governs Paper 16 claims and keeps window receipts separate from global continuum closure.

Admitted Claims

Admit a claim when it provides:

```
local chart state
anneal steps
rest-state result
edge-residue value
window depth
open/global boundary, if any
```

Rejected Promotions

Reject by default:

```
power-of-ten window -> continuum solution
local residue -> closed global correction sum
oracle/local nth-bit pass ->  $O(\log N)$  extractor
```

Hidden-Guess Rule

Every diagnostic must hide the answer until classification:

```
closed local
open global
invalid overclaim
bounded/oracle evidence
```

Linked Receipt

Current receipt:

```
production/formal-papers/CQE-paper-16/continuum_edge_residuals_receipt.json
```

Current verdict:

```
local edge-residual windowing: pass
global continuum collapse: open
```

Boundary Failure

The contract fails if a later paper says Paper 16 solves the continuum limit.

The contract passes if a later paper says Paper 16 supplies local residual windows that expose the still-open continuum-depth problem.

Conclusion

Paper 16.50 keeps the local instrument useful and the global obligation visible.